Post Translational Modification

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Objectives

- Define post translational mechanism
- Describe the different chemical reaction for mature functional protein
- Justify the role of post translational mechanism in protein synthesis

Definition

- Is the chemical modification of a protein after its translation. It is one of the later steps in protein biosynthesis for many proteins.
- Covalently modified → attachment to ribosome
- Initiated → translation
- Removal of part of translated sequences, covalent addition, of chemical groups

Types

- Trimming
- Covalent alterations → Phosphorylation, Glycosylation, Hydroxylation, carboxylation
- Protein degradation

Trimming

- Large precursor molecules → not functionally active
- Endoproteases → remove portion of protein chain → active molecule
- ER, Golgi apparatus, secretory vesicle →Insulin
- Collagen → after secretion → cleaved
- Zymogens → inactive precursors → site of action → cleaved → activated

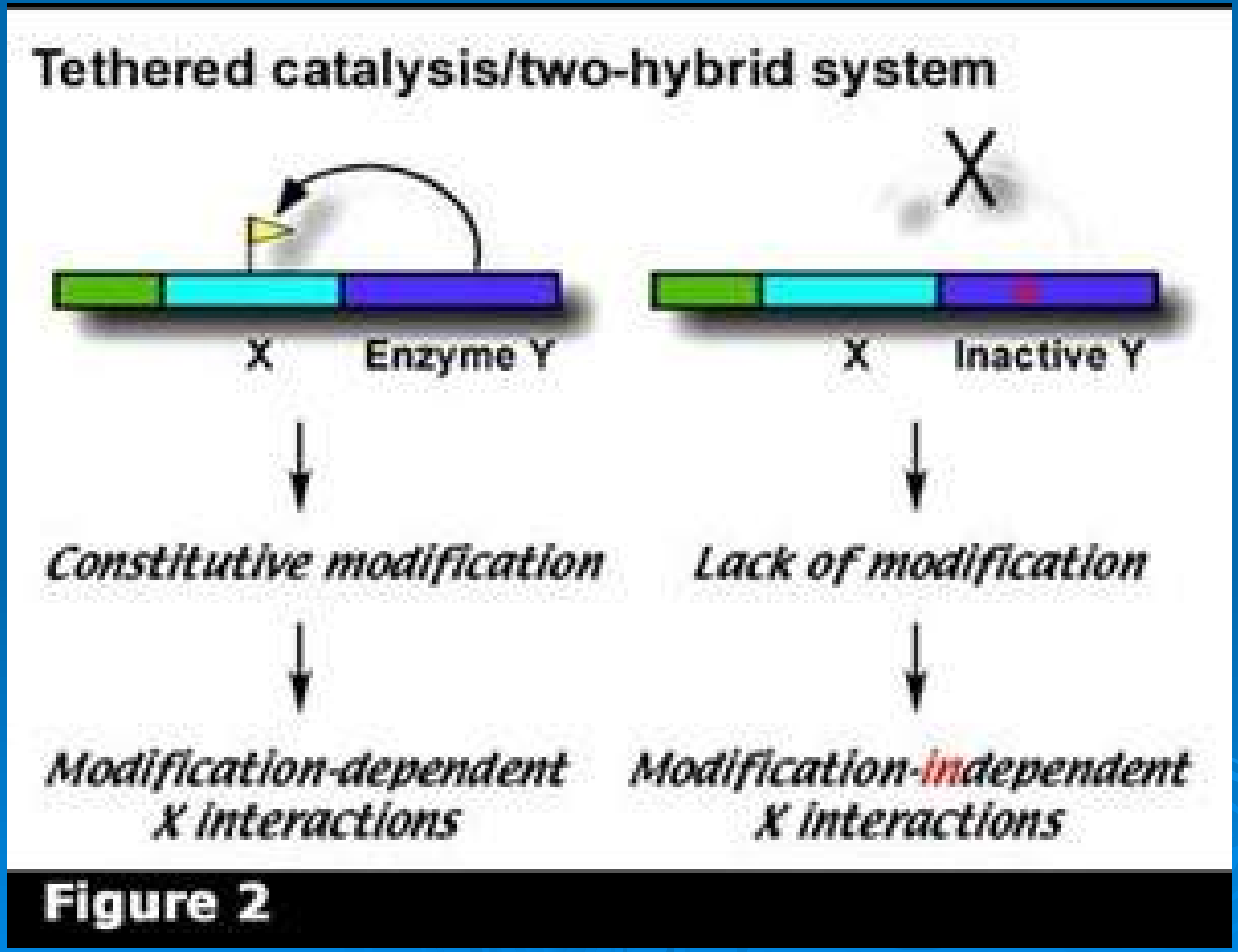

Tethered catalysis/two-hybrid system
X
X
Enzyme Y
X
Inactive Y
Constitutive modification
Lack of modification
Modification-dependent X interactions
Modification-independent X interactions

Figure 2

Covalent Alteration

- Enzymatic and structural → chemical groups
- Phosphorylation → OH of serine, threonine, tyrosine → catalyzed by protein kinases, phosphatases → inctreases or decreases the functional activity of the protein

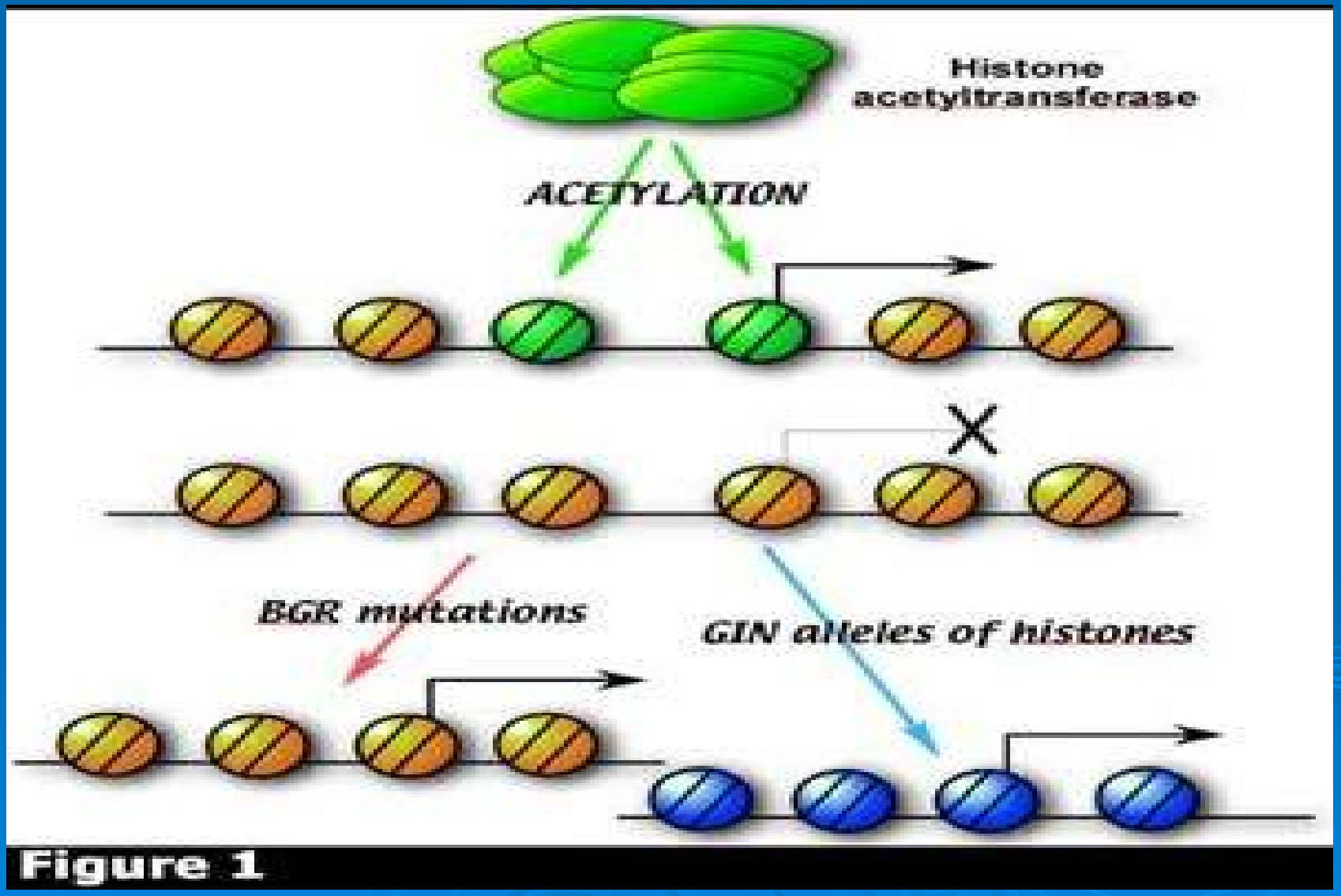

Histone
acetyltransferase
ACETYLATION
BGR mutations
GIN alleles of histones

Figure 1

- Glycosylation → protein → plasma membrane or lysosome → CHO chain attached to serine or threonine OH gp (O linked) → amide N of aspargine (N linked)
- ER, GA

- Hydroxylation → Proline, lysine residues → α chains of collagen → hydroxylated
- ER

- Other Covalent modification → carboxylation
- Carboxyl added → glutamate → vitamin K dependent →γ carboxyglutamate residus→ blood clotting protein
- Attachment of lipids → farnesyl gp → anchor proteins in membranes

Protein Degradation

- Defective proteins → proteasome
- Complex,ATP dependent proteolytic system → cytosol